

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Department: CSE - AIDS	Course Code: DSA0604
Course Name: Data Handling and Visualization	

DATA VISUALIZATION — SET 11 TO 15

SET 11: PRODUCT CATEGORY ANALYSIS

Dataset: Product Categories

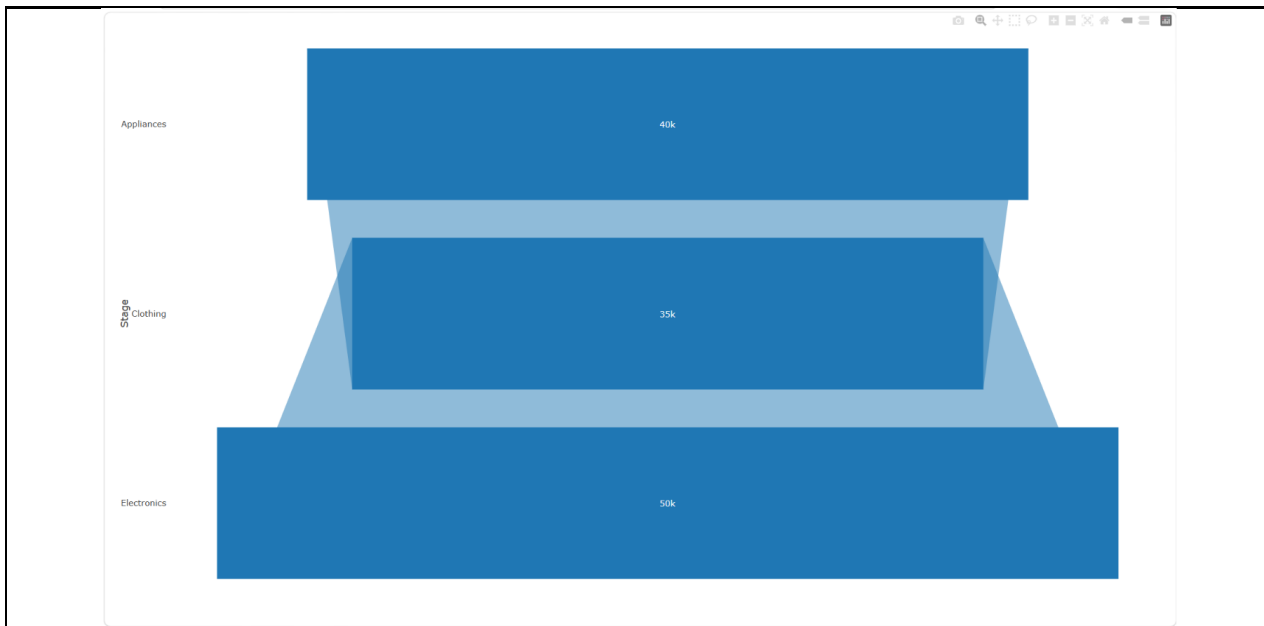
Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

Q1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

CODE:

```
install.packages("plotly")
library(plotly)
category_data <- data.frame(
  Stage = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
plot_ly(
  category_data,
  type = "funnel",
  y = ~Stage,
  x = ~Sales
)
```

OUTPUT:



Q2. Build a table to display the sales data for each product category. Label the table elements.

CODE:

```
category_data <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
print(category_data)
```

OUTPUT:

```
> category_data <- data.frame(  
+   Category = c("Electronics", "Clothing", "Appliances"),  
+   Sales = c(50000, 35000, 40000)  
+ )  
>  
> print(category_data)  
  Category Sales  
1 Electronics 50000  
2   Clothing 35000  
3 Appliances 40000  
> |
```

Q3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

CODE:

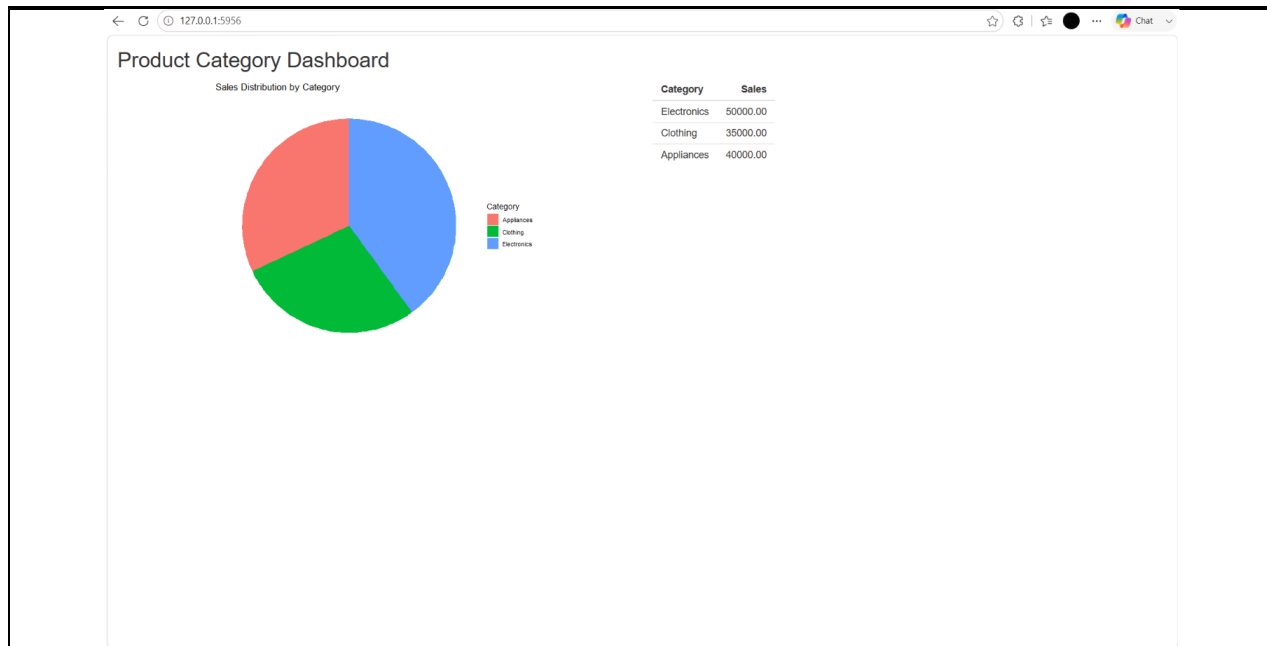
```
library(shiny)  
library(ggplot2)  
category_data <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)
```

```

ui <- fluidPage(
  titlePanel("Product Category Dashboard"),
  fluidRow(
    column(6, plotOutput("piePlot")),
    column(6, tableOutput("salesTable"))
  )
)
server <- function(input, output) {
  output$piePlot <- renderPlot({
    ggplot(category_data, aes(x = "", y = Sales, fill = Category)) +
    geom_bar(stat = "identity", width = 1) +
    coord_polar("y") +
    labs(title = "Sales Distribution by Category") +
    theme_void()
  })
  output$salesTable <- renderTable({
    category_data
  })
}
shinyApp(ui, server)

```

OUTPUT:



Q4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

CODE:

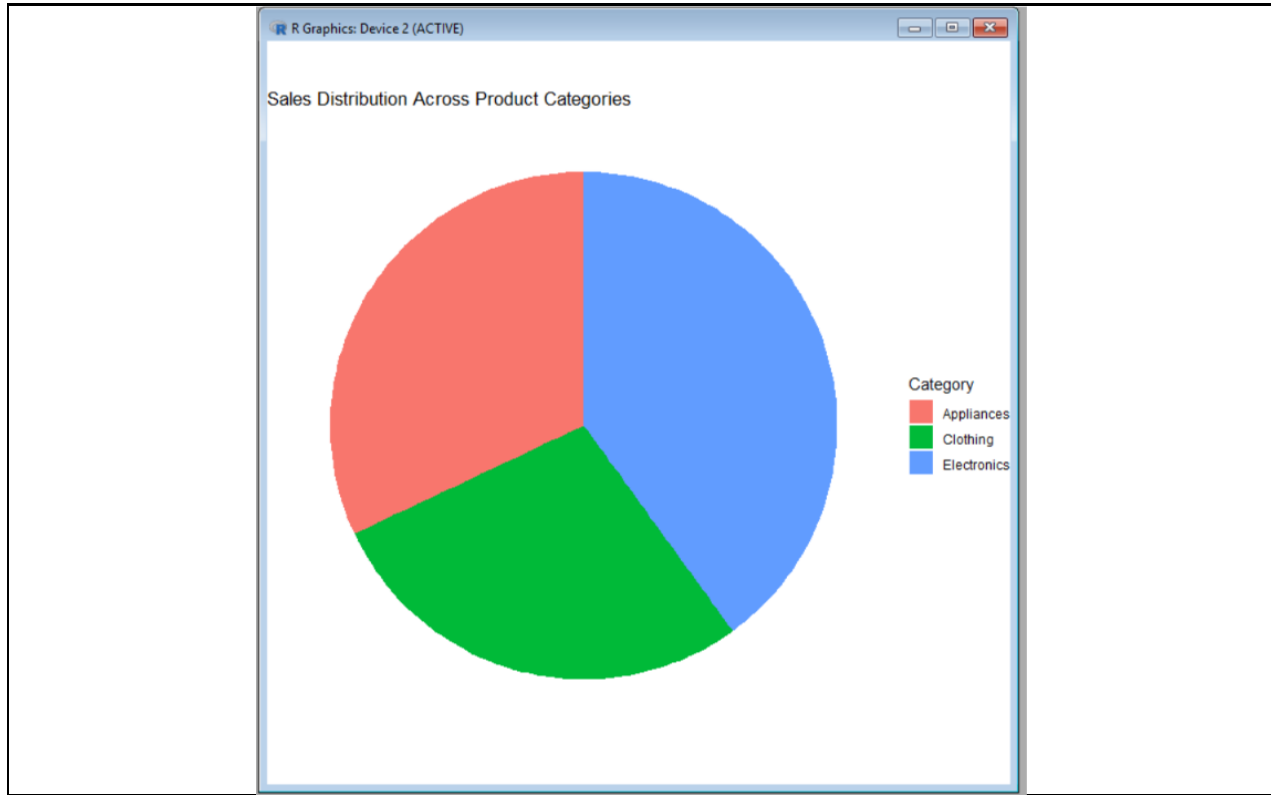
```

library(ggplot2)
category_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
ggplot(category_data, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
    title = "Sales Distribution Across Product Categories",
    fill = "Category"
  )

```

```
) +  
theme_void()
```

OUTPUT:



SET 12: WEBSITE TRAFFIC ANALYSIS

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

Q1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

CODE:

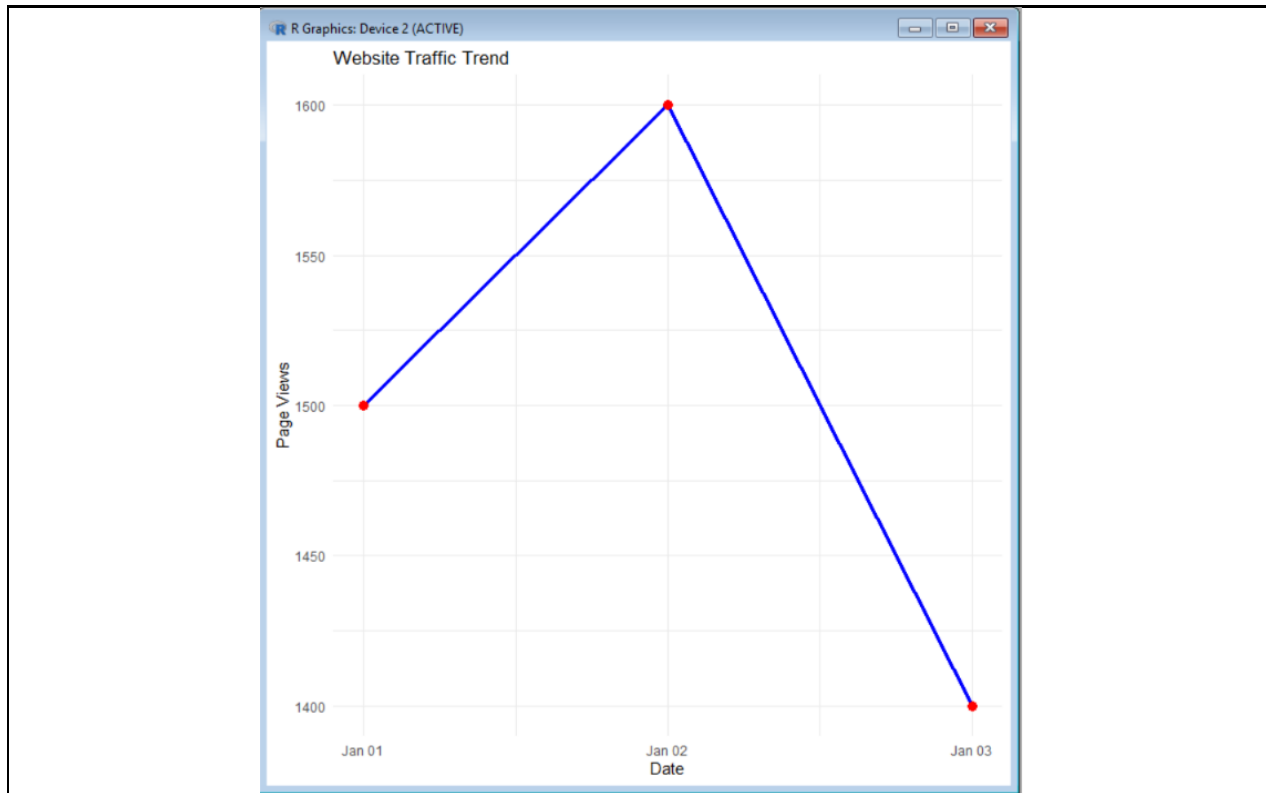
```
library(ggplot2)  
website_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c(2.3, 2.7, 2.0)  
)  
ggplot(website_data, aes(x = Date, y = Page_Views, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views Trend",  
    xlab = "Date",  
    ylab = "Page Views",  
    theme = theme_minimal()
```

```

title = "Website Traffic Trend",
x = "Date",
y = "Page Views"
) +
theme_minimal()

```

OUTPUT:



Q2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

CODE:

```

library(ggplot2)
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  Click_Through_Rate = c(2.3, 2.7, 2.0)
)
ggplot(website_data, aes(x = factor(Date), y = Click_Through_Rate, fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Click-Through Rate by Date",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

```

OUTPUT:



Q3. Build a table to show the website traffic data for each day. Label the table elements.

CODE:

```
website_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c(2.3, 2.7, 2.0)  
)  
print(website_data)
```

OUTPUT:

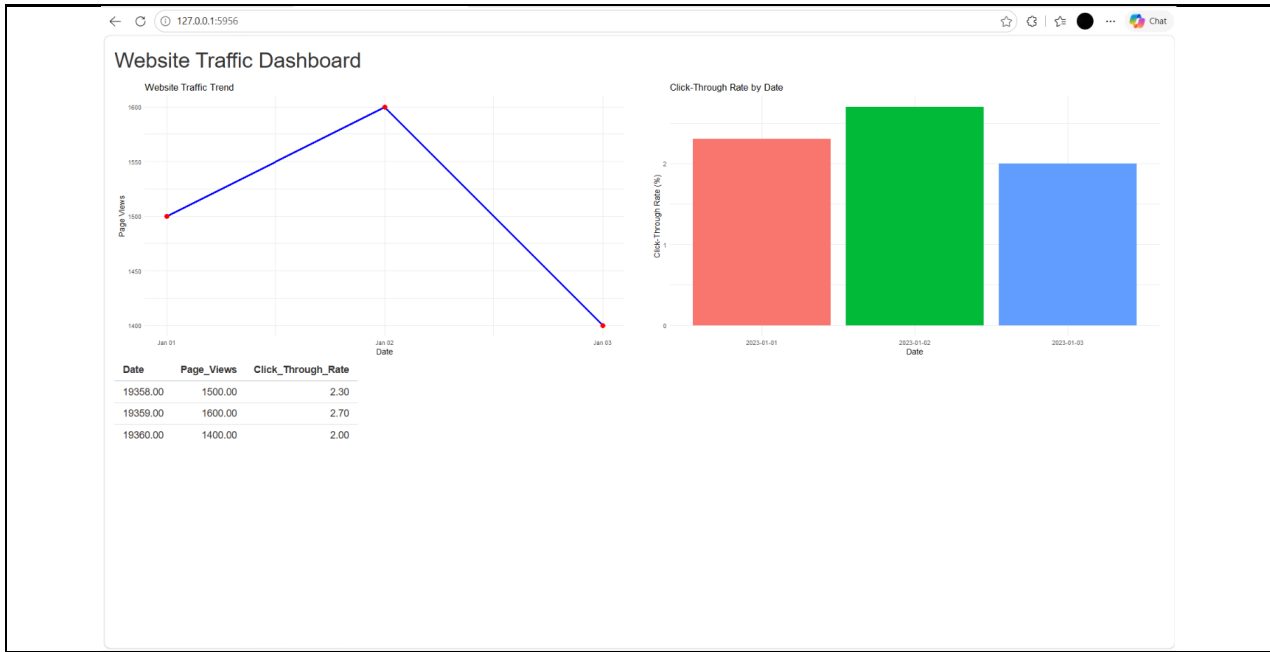
Date	Page_Views	Click_Through_Rate
19358.00	1500.00	2.30
19359.00	1600.00	2.70
19360.00	1400.00	2.00

Q4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

CODE:

```
library(shiny)
library(ggplot2)
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Page_Views = c(1500, 1600, 1400),
  Click_Through_Rate = c(2.3, 2.7, 2.0)
)
ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  fluidRow(
    column(6, plotOutput("linePlot")),
    column(6, plotOutput("barPlot"))
  ),
  fluidRow(
    column(12, tableOutput("trafficTable"))
  )
)
server <- function(input, output) {
  output$linePlot <- renderPlot({
    ggplot(website_data, aes(x = Date, y = Page_Views, group = 1)) +
      geom_line(color = "blue", linewidth = 1.2) +
      geom_point(color = "red", size = 3) +
      labs(
        title = "Website Traffic Trend",
        x = "Date",
        y = "Page Views"
      ) +
      theme_minimal()
  })
  output$barPlot <- renderPlot({
    ggplot(website_data, aes(x = factor(Date), y = Click_Through_Rate, fill = factor(Date))) +
      geom_bar(stat = "identity") +
      labs(
        title = "Click-Through Rate by Date",
        x = "Date",
        y = "Click-Through Rate (%)"
      ) +
      theme_minimal() +
      theme(legend.position = "none")
  })
  output$trafficTable <- renderTable({
    website_data
  })
}
shinyApp(ui, server)
```

OUTPUT:



SET 13: GEOGRAPHIC DATA ANALYSIS

Dataset: Geographic Data

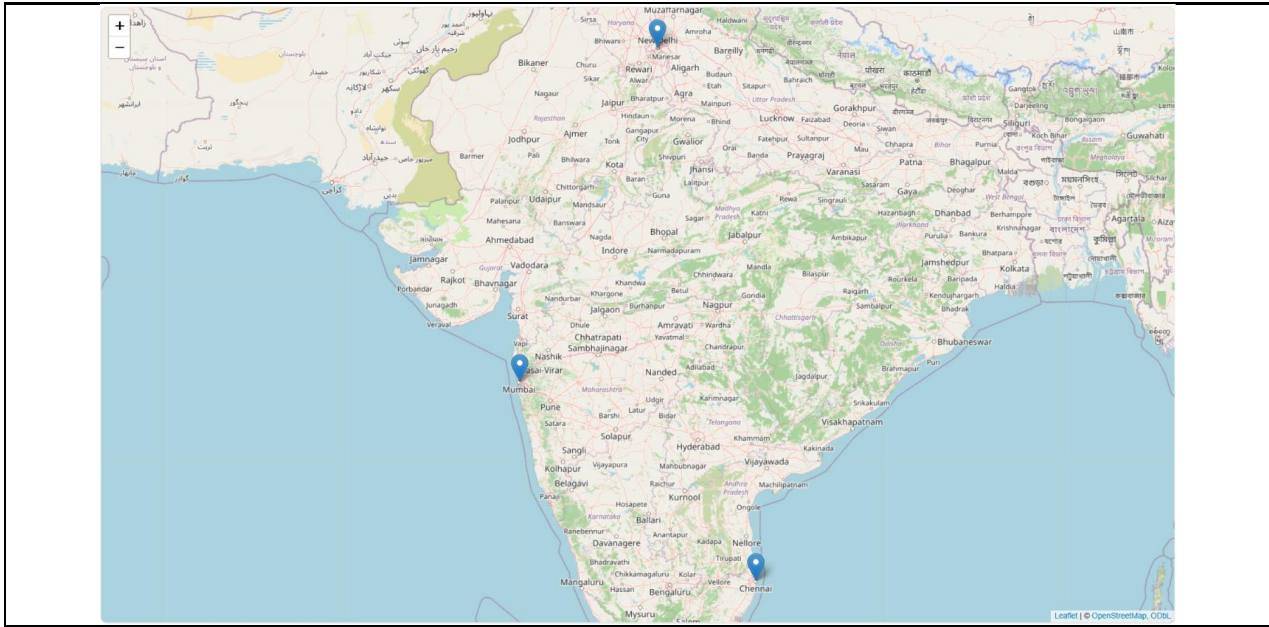
City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Q1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

CODE:

```
install.packages("leaflet")
library(leaflet)
city_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Latitude = c(13.08, 28.61, 19.07),
  Longitude = c(80.27, 77.20, 72.88),
  Population = c(500000, 700000, 600000)
)
leaflet(city_data) %>%
  addTiles() %>%
  addMarkers(
    lng = ~Longitude,
    lat = ~Latitude,
    popup = ~paste(City, "<br>Population:", Population)
  )
```

OUTPUT:

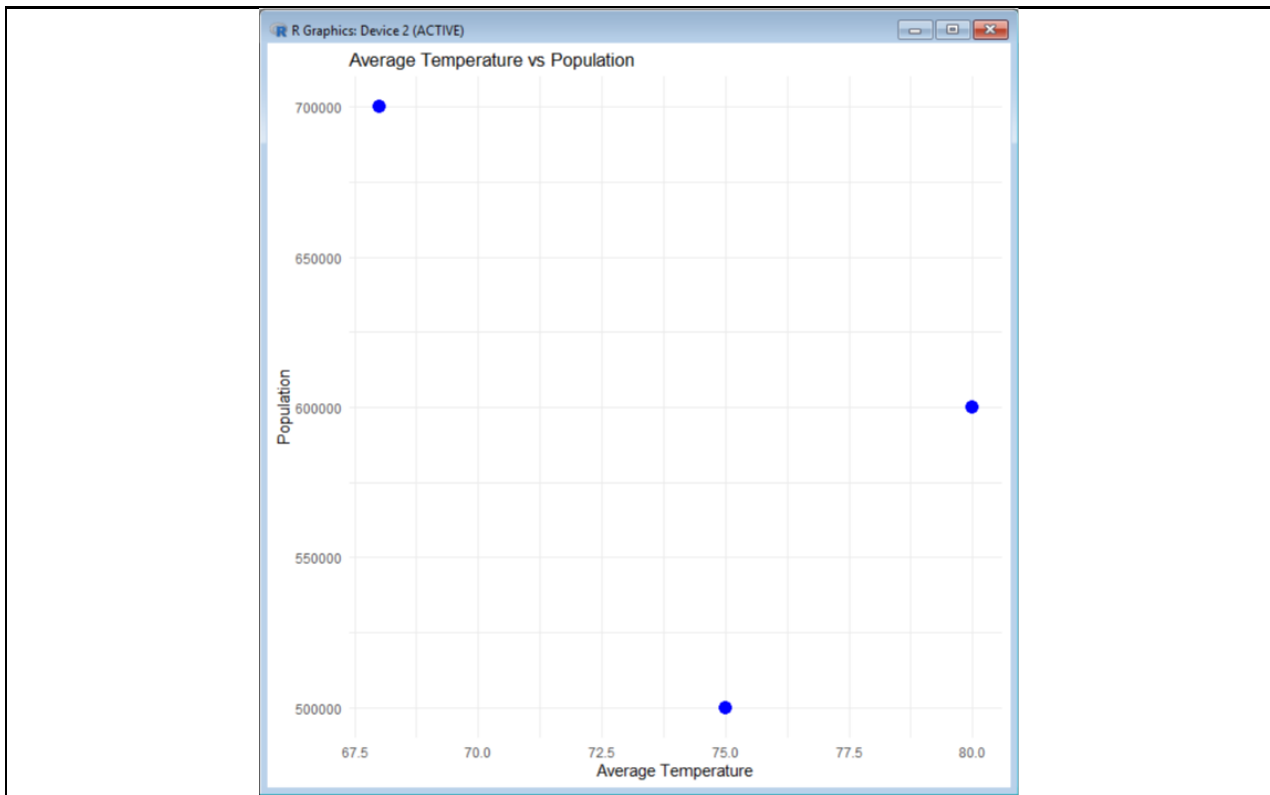


Q2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

CODE:

```
library(ggplot2)
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
ggplot(geo_data, aes(x = Avg_Temperature, y = Population)) +
  geom_point(size = 4, color = "blue") +
  labs(
    title = "Average Temperature vs Population",
    x = "Average Temperature",
    y = "Population"
  ) +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to display the geographic data for each city. Label the table elements.

CODE:

```
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
print(geo_data)
```

OUTPUT:

City	Population	Avg_Temperature	Elevation
City A	500000.00	75.00	1000.00
City B	700000.00	68.00	800.00
City C	600000.00	80.00	1200.00

Q4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

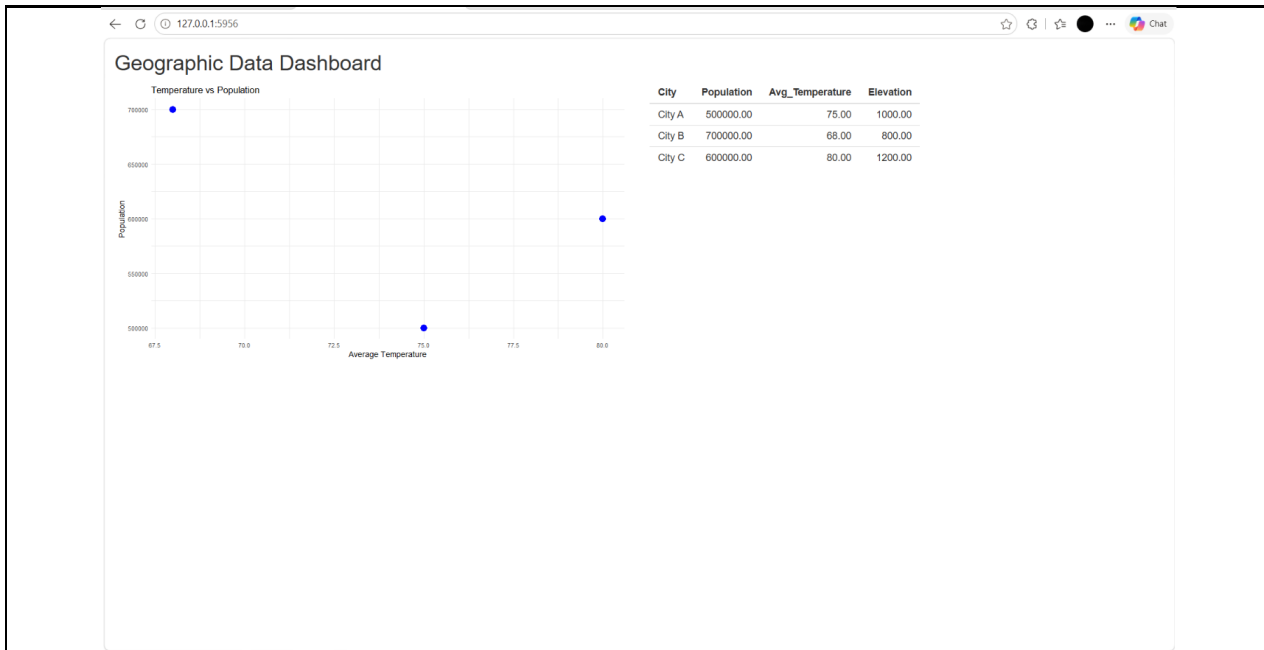
CODE:

```

library(shiny)
library(ggplot2)
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
ui <- fluidPage(
  titlePanel("Geographic Data Dashboard"),
  fluidRow(
    column(6, plotOutput("scatterPlot")),
    column(6, tableOutput("geoTable"))
  )
)
server <- function(input, output){
  output$scatterPlot <- renderPlot({
    ggplot(geo_data, aes(x = Avg_Temperature, y = Population)) +
    geom_point(size = 4, color = "blue") +
    labs(
      title = "Temperature vs Population",
      x = "Average Temperature",
      y = "Population"
    ) +
    theme_minimal()
  })
  output$geoTable <- renderTable({
    geo_data
  })
}
shinyApp(ui, server)

```

OUTPUT:



SET 14: SURVEY RESULTS ANALYSIS

Dataset: Survey Results

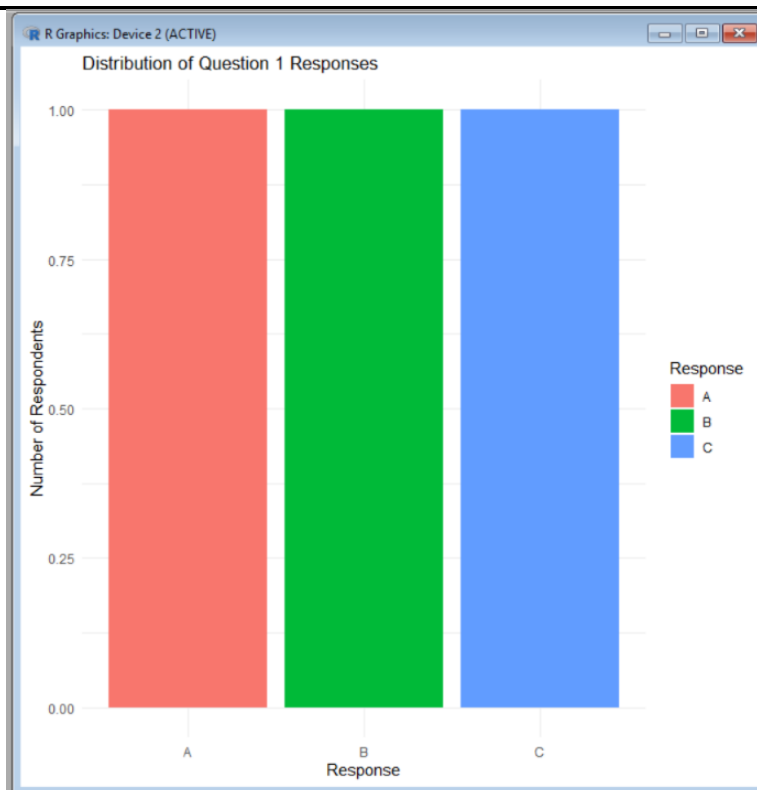
Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

CODE:

```
library(ggplot2)
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Response", "Count")
ggplot(q1_data, aes(x = Response, y = Count, fill = Response)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Question 1 Responses",
    x = "Response",
    y = "Number of Respondents"
  ) +
  theme_minimal()
```

OUTPUT:

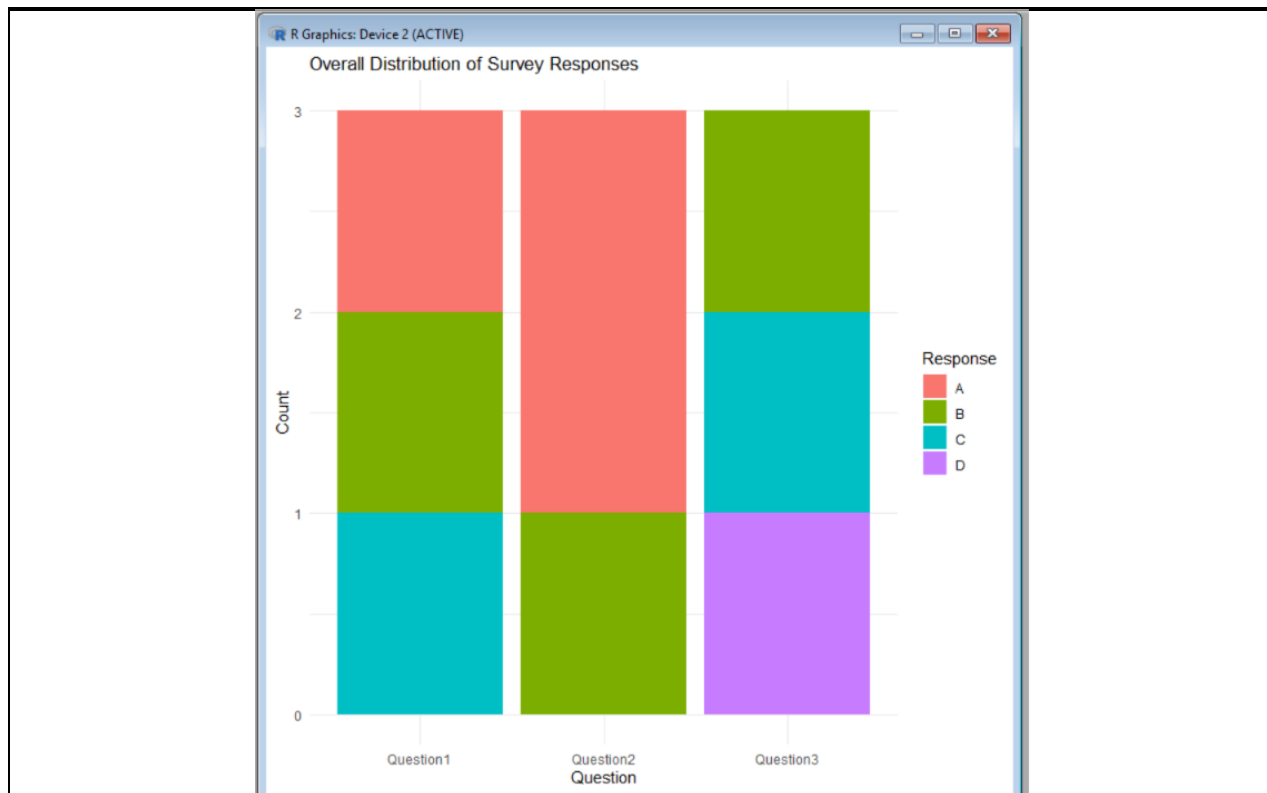


Q2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

CODE:

```
library(ggplot2)
library(tidyr)
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
survey_long <- pivot_longer(
  survey_data,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long, aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Question",
    y = "Count",
    fill = "Response"
  ) +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to show the survey response data for each respondent. Label the table elements.

CODE:

```

survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
print(survey_data)

```

OUTPUT:

Respondent	Question1	Question2	Question3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

Q4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

CODE:

```

library(shiny)
library(ggplot2)
library(tidyr)
survey_data <- data.frame(
  Respondent = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Response", "Count")
survey_long <- pivot_longer(
  survey_data,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)
ui <- fluidPage(
  titlePanel("Survey Results Dashboard"),
  fluidRow(
    column(6, plotOutput("groupedBar")),
    column(6, plotOutput("stackedBar"))
  ),
  fluidRow(
    column(12, tableOutput("surveyTable"))
  )
)
server <- function(input, output) {
  output$groupedBar <- renderPlot({
    ggplot(q1_data, aes(x = Response, y = Count, fill = Response)) +

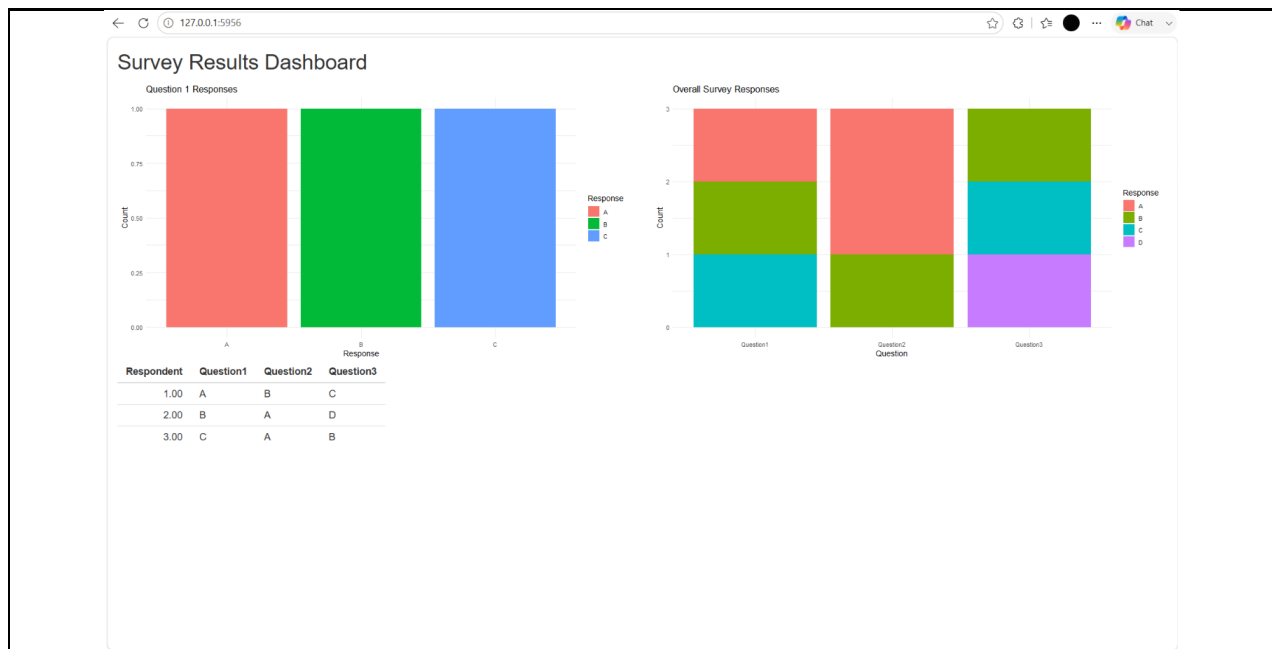
```

```

geom_bar(stat = "identity") +
labs(
title = "Question 1 Responses",
x = "Response",
y = "Count"
) +
theme_minimal()
})
output$stackedBar <- renderPlot({
ggplot(survey_long, aes(x = Question, fill = Response)) +
geom_bar() +
labs(
title = "Overall Survey Responses",
x = "Question",
y = "Count"
) +
theme_minimal()
})
output$surveyTable <- renderTable({
survey_data
})
}
shinyApp(ui, server)

```

OUTPUT:



SET 15: PRODUCT SALES ANALYSIS

Dataset: Monthly Product Sales

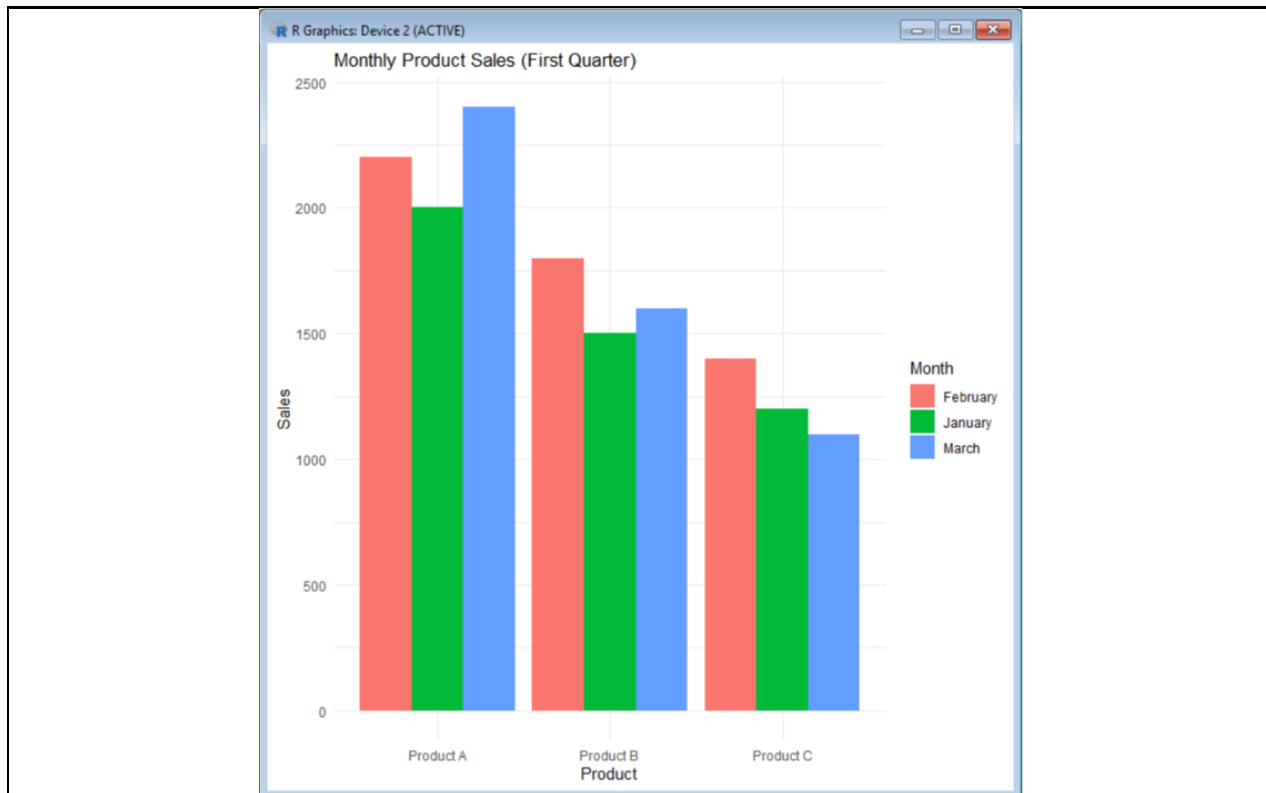
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Q1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

CODE:

```
library(ggplot2)
library(tidyr)
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_long <- pivot_longer(
  sales_data,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long, aes(x = Product, y = Sales, fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Monthly Product Sales (First Quarter)",
    x = "Product",
    y = "Sales"
  ) +
  theme_minimal()
```

OUTPUT:

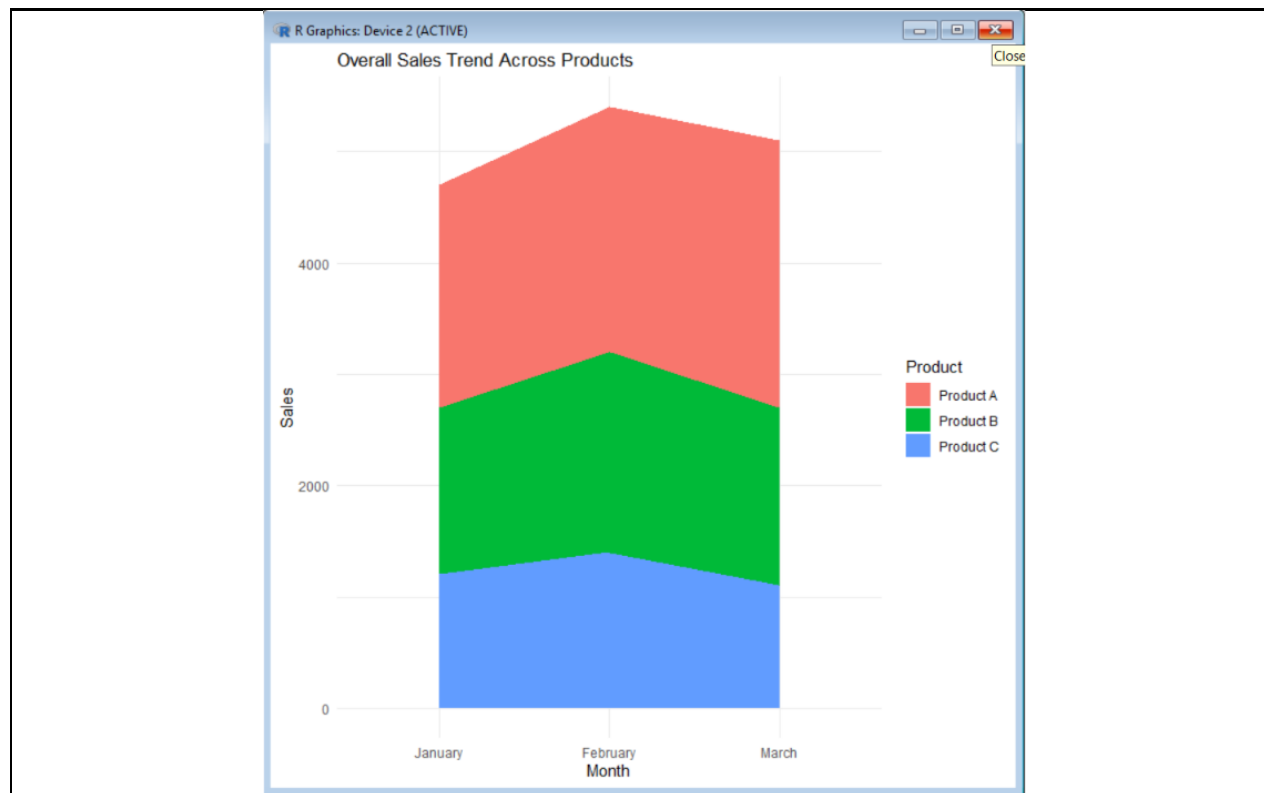


Q2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

CODE:

```
library(ggplot2)
library(tidyr)
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_long <- pivot_longer(
  sales_data,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)
sales_summary <- aggregate(Sales ~ Month + Product, sales_long, sum)
sales_summary$Month <- factor(
  sales_summary$Month,
  levels = c("January", "February", "March")
)
ggplot(sales_summary, aes(x = Month, y = Sales, fill = Product, group = Product)) +
  geom_area(position = "stack") +
  labs(
    title = "Overall Sales Trend Across Products",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```

OUTPUT:



Q3. Build a table to show the monthly sales data for each product. Label the table elements.

CODE:

```
sales_data <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  January_Sales = c(2000, 1500, 1200),  
  February_Sales = c(2200, 1800, 1400),  
  March_Sales = c(2400, 1600, 1100)  
)  
print(sales_data)
```

OUTPUT:

Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00

Q4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

CODE:

```
library(shiny)  
library(ggplot2)  
library(tidyr)  
sales_data <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)  
sales_long <- pivot_longer(  
  sales_data,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)  
ui <- fluidPage(  
  titlePanel("Product Sales Dashboard"),  
  fluidRow(  
    column(6, plotOutput("barPlot")),  
    column(6, plotOutput("areaPlot"))  
  ),  
  fluidRow(  
    column(12, tableOutput("salesTable"))  
  )  
)  
server <- function(input, output) {  
  output$barPlot <- renderPlot({  
    ggplot(sales_long, aes(x = Product, y = Sales, fill = Month)) +
```

```

geom_bar(stat = "identity", position = "dodge") +
labs(
  title = "Monthly Product Sales",
  x = "Product",
  y = "Sales"
) +
theme_minimal()
})
output$areaPlot <- renderPlot({
  sales_summary <- aggregate(Sales ~ Month + Product, sales_long, sum)
  sales_summary$Month <- factor(
    sales_summary$Month,
    levels = c("January", "February", "March")
  )
  ggplot(sales_summary, aes(x = Month, y = Sales, fill = Product, group = Product)) +
  geom_area(position = "stack") +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
})
output$salesTable <- renderTable({
  sales_data
})
}
shinyApp(ui, server)

```

OUTPUT:

